6.1 Perpendicular and Angle Bisectors

Name _____ Period ____ Date _____

Geometry 6.1 · Student sheet

I can…

- use the Perpendicular Bisector Theorem and its converse to find lengths.
- use the Angle Bisector Theorem and its converse to find lengths and angle measures.
- write an equation of a perpendicular bisector.

Vocabulary · fill in the missing word

Equidistant — the same _____ from two or more objects

Perpendicular bisector — a line, segment, or ray that is perpendicular to a segment at its _____

Distance from a point to a line — the length of the _____ segment from the point to the line

Angle bisector — a ray that divides an angle into two _____ angles

Part 1 · Patty paper discovery

Fold 1. Draw a segment $\overline{AB}$. Fold A onto B. The crease is the perpendicular bisector. Mark three points on the crease and measure.

Point on the crease	Distance to A	Distance to B

What I notice:

Fold 2. Draw an angle. Fold one side onto the other. The crease is the angle bisector. Mark a point on it and measure the **perpendicular** distance to each side.

Point on the crease	Distance to side 1	Distance to side 2

What I notice:

Part 2 · Notes: fill in the blanks

Perpendicular Bisector Theorem: If a point is on the perpendicular bisector of a segment, then it is _____ from the endpoints of the segment.

Converse: If a point is equidistant from the endpoints of a segment, then it is on the _____ _____ of the segment.

Angle Bisector Theorem: If a point is on the bisector of an angle, then it is equidistant from the two _____ of the angle.

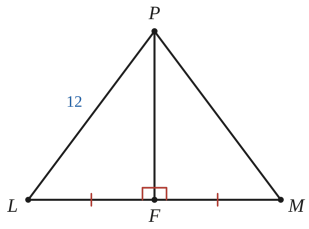
Converse: If a point is in the interior of an angle and is equidistant from the two sides, then it lies on the _____ of the angle.

The distance from a point to a line is the length of the _____ segment from the point to the line.

Part 3 · Examples (work them with the class)

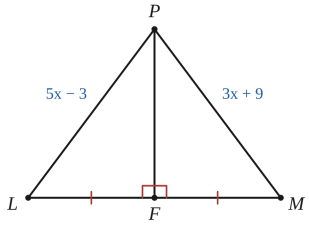
Example 1

$\overline{PF}$ is the perpendicular bisector of $\overline{LM}$. Find PM.



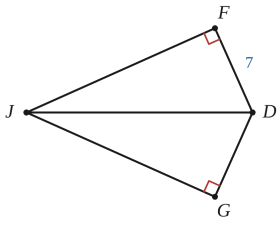
Example 2

$\overline{PF}$ is the perpendicular bisector of $\overline{LM}$. Find PL.



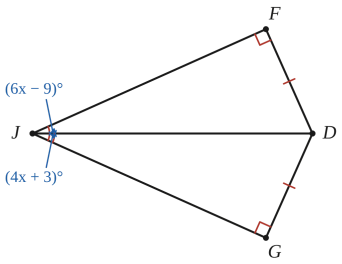
Example 3

$\overline{JD}$ bisects $\angle FJG$. Find DG.



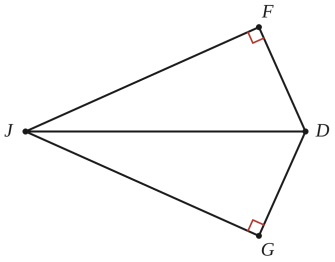
Example 4

Find m∠FJG.



Example 5

Can you conclude that $\overline{JD}$ bisects ∠FJG? Explain.



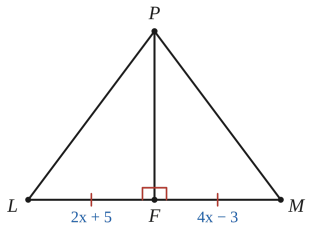
Example 6

Write an equation of the perpendicular bisector of the segment with endpoints (−2, 3) and (4, 1).

Part 4 · Partner practice

Partner 1

$\overline{PF}$ is the perpendicular bisector of $\overline{LM}$. Find LM.



Partner 2

$\overline{JD}$ bisects $\angle FJG$. Find DF.

